A Data Visualization Framework during Pandemic using the Density-Based Spatial Clustering with Noise (DBSCAN) Machine Learning Model

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2021 IEEE 11th International Conference on System Engineering and Technology (ICSET) | 978-1-6654-3766-0/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICSET53708.2021.9612563

***Abstract*—Big data technologies have become an important part in our life, especially during the pandemic. These technologies can be used to collect, analyse, process, and interpret the collected data in order to produce a useful information or knowledge. In fact, we are depending on the information extracted from the large amount of data collected daily from mobile applications. One of the examples of the application that has been used in Malaysia is MySejahtera which provides useful information on the spread of the pandemic. The data can be clustered using machine learning models such as clustering algorithm. Therefore, in this project, we propose a framework that will be useful to monitor the information about COVID-19 and visualizing the information with a machine learning model. The data visualization can help with data interpretation and improving how we can manage the spread of the virus. This project was also implemented using a modified waterfall which allows the developer to return to the previous phase in order to make some modifications before the final product can be used by users. This project used a Python approach to develop a dashboard. A Density-Based Spatial Clustering with Noise algorithm was chosen for the data classification of the countries based on its number of cases and number of deaths.**

***Index Terms*—Big data, COVID-19, machine learning, clustering, data visualization, DBSCAN, dashboard**

I. Introduction

On December 31, 2019, the Coronavirus disease or commonly known as COVID-19, was discovered in Wuhan City, Hubei Province, China. Fever and cough are some of the symptoms that an infected individual may experience. The World Health Organization (WHO) officially announced COVID-19 a pandemic in March 2020. Many governments have put their countries on hold, with the goal of limiting movements to help stop the virus from spreading further [1]. Persons can become infected when they come into close contact with infected people, according to the WHO. The virus can enter their mouth, nose, or eyes by respiratory droplets. Close-contact situations, crowded environments, and tight spaces with insufficient ventilation make transmissions simpler [2].

During a pandemic, people must wear masks, check their temperature before accessing public places, and keep a close eye on their social distance. All of the information can be gathered using a mobile application. In Malaysia, for example, data can be collected using the MySejahtera application. This is because the mobile application collects and stores a vast quantity of data, and it must be examined and processed in order to track our movements and prevent the virus from spreading. Another issue to be cautious of is the contradictory and twisted information available on the internet. It can also be seen as a public health risk. Misleading information can smear the reputations of well-known doctors and cause people to doubt the efficacy of a lab-tested vaccine [3]. As a result, it is critical that citizens have access to the most up-to-date and accurate information available.

The project's three objectives are as the followings: 1) to design a dashboard that can visualize the current COVID-19 situation, 2) to develop a dashboard with machine learning model: a density-based clustering algorithm, and 3) to test the functionality and usability of the dashboard.

This project focuses on visualizing the situation of pandemic all around the world using machine learning model. The targeted users are web users from all around the globe, despite their background and age. The dataset used is a dataset of secondary data from the GitHub website.

The significance of this project is to visualize the data of COVID-19 cases all around the world. The data visualization can be clustered using machine learning model such as clustering algorithm. The data can also be used to monitor and understand how the pandemic spreads. The application's users can monitor the updates on the current statistics of COVID-19. This will help avoid fake news that may lead to misinformation which consequently may induce panic. Furthermore, everyone will profit from this application, including ordinary citizens, public healthcare providers, and legislators. An economist, for example, will keep an eye on the data and advises the government on how to improve citizens' protection through healthcare [4].

The rest of the paper is organized as the following: The literature review and methodology of this project are discussed in Sections 2 and 3, respectively. The limitation, future work, and conclusion of this project are presented in Section 4.

II. Literature review

A. Big data

Big data is a technology that can be used to process and extract information or knowledge from a large amount of data [5]. This useful information can be assessed by online users through websites or mobile applications. Patterns, trends, relationships, and differences can all aid in the dissemination of information and knowledge regarding the virus's spread and management [6].

Mobile applications can also be used to capture vast amounts of data. In Malaysia, for example, statistics about COVID-19 can be obtained through MySejahtera, a government-developed application [7]. For the purpose of registration, MySejahtera uses QR code where the information will be kept in the record for 30 days. These data can be utilized by healthcare workers or the government to take prompt actions including contact tracing to stop the virus from spreading [6].

As a result, big data technologies can be employed to speed up the process of processing the application's collected data. The information can be utilized to devise strategies for combating the virus and to learn about the pandemics after-effects.

Fig. 1 depicts a possible application that might be used in conjunction with big data technology. For example, data visualization can aid in determining the number of COVID-19 instances and categorizing them according to their location. Therefore, it would be easier for healthcare facilities to detect and track the infected individuals and those who are a close contact with the infected individuals. At the same time, the data can be used by policy-makers to provide the number of cares needed and places for quarantine.

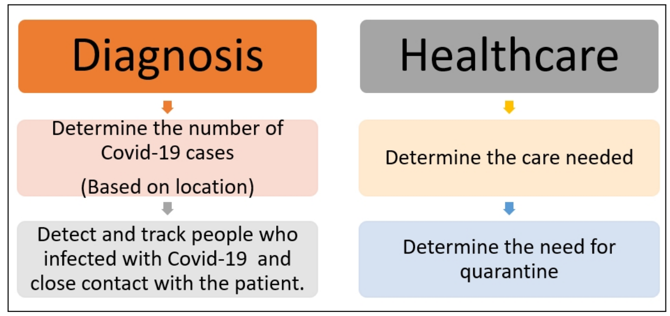

Fig. 1. Potential application that can be integrated with big data technologies.

B. Data visualization

The technique of visualising data is known as data visualisation which provides relevant data and assists users in better understanding the data [8]. Histograms, area charts, fever charts, info-graphics, and heat map visualisations can all be utilized to analyse data. For example, Fig. 2 shows the heat map of COVID-19 cases in Kuala Lumpur, Malaysia. The red colour shows a high number COVID-19 cases in Kuala Lumpur (Source: *https://www.arcgis.com/*).

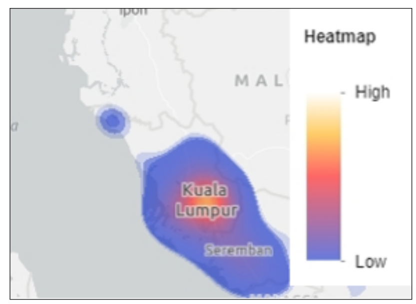

Fig. 2. Heat map of COVID-19 cases in Kuala Lumpur, Malaysia.

C. Machine learning

Computer science has a field called machine learning. Machine learning is used to improve performance based on a sample of data or previous experience. In machine learning, there are two categories: 1) supervised and 2) unsupervised. The unsupervised learning can be used to detect similarities in data points and then group the data based on the similarities, whereas the supervised learning can be used to find patterns in the data [9].

The unsupervised learning can be utilized to categorize data based on similarities, and it can be used to group data. In data visualization, the model can be utilized to cluster data. K-Means is one of the algorithms in unsupervised learning, (2) Mini-Batch K-Means, and (3) Density-based spatial clustering of applications with noise (DBSCAN).

D. Clustering algorithms

This section will discuss the clustering algorithms that can be used for data visualization: (1) K-Means, (2) Mini-Batch K-Means, and (3) DBSCAN.

Clustering is a technique for constructing groupings of vast amounts of data that has been utilized in a variety of domains, including machine learning [10]. There are various algorithms to choose from when it comes to clustering algorithms. Each algorithm follows its own set of rules. This set of principles is used to determine the formation of a cluster. As a result, this part will concentrate on a few different types of clustering algorithms, how they function, how they may be applied in the desired circumstance, and which clustering algorithm is best for the suggested application.

1) K-Means: One of the most widely used clustering algorithms is K-Means. It has a number of features, including the ability to expand to big datasets, guarantee convergence, and rapidly adapt to new examples [11]. The drawback is that the users must manually select the K, which is the optimal. This method of clustering is also ineffective in instances where clusters of varying sizes exist. Aside from that, the datasets for this technique will have to be stored in main memory. As a result, as the size of the dataset being evaluated grows larger, it becomes less appealing.

2) Mini-Batch K-Means: Mini-Batch K-Means is a wonderful alternative to K-Means. The algorithm can be used when working with a huge dataset. This is due to the fact that it employs small batches of data that were chosen at random. The batches with fixed sizes are then saved in memory. The algorithm's learning rate is updated by employing the gradient descent approach to update its centroid. It is used to achieve rapid convergence. In each iteration where a random sample from the data set is chosen, the centroids are periodically updated before convergence [11]. Mini-Batch K-Means is quicker than K-Means. However, the outcomes may vary slightly. It does not come without a cost, despite the fact that it minimizes the amount of distance computations for each iteration. To compensate for the increased number of computations, the cluster's quality must be sacrificed.

3) DBSCAN algorithm: Density-Based Spatial Clustering of Applications with Noise (DBSCAN) falls under the category of density-based clustering algorithm. It varies from the others in that the clusters are based on the density of the area rather than a predefined size or segmentation of the data. The Euclidean distance is used to group points that are close to each other in this procedure. The minimum number of points can also be changed depending on the space; hence, there will be more noise. This approach does not function well with high-dimensional data with more features than the sample size. Data for visualization, for example, can yield hundreds of samples [12].

E. Discussion

The amount of data that the system will have to cope with must be taken into account when selecting an algorithm. Aside from that, the speed with which the algorithm can deliver a result is important because time is of the essence in the fight against the epidemic. The DBSCAN algorithm has been determined to be the most suitable clustering algorithm after a thorough comparison. The number of clusters must be specified in advance, K-Means cannot be employed. In addition, the K-Means algorithm assumes that each cluster has a convex form. Even though Mini-Batch K-Means is marginally faster than K-Means in terms of speed, the difference is not substantial. This technique minimizes the amount of distance computations per iteration at the expense of reduced cluster quality, which is not an option because, even though time is of the essence, information must be precise and not of low quality when fighting a highly spreading virus. To put it in context, the DBSCAN algorithm is around four times faster than the K-Means technique. Its merging procedure is also irreversible, which might be a significant disadvantage when working with noisy or high-dimensional data.

DBSCAN is the most appropriate algorithm overall since this virus is highly contagious in congested areas. DBSCAN is the only clustering method that produces clusters based on density among all of the algorithms listed above. Other than that, the clusters cannot be restricted to a certain shape or size.

III. Methodology

This project was implemented using a modified waterfall methodology [13]. There are seven phases in the methodology. However, the developer only used four phases of the modified waterfall for this project. The four phases are (i) Requirement gathering and analysis, (ii) Design, (iii) Implementation, and (iv) Testing.

Fig. 3 shows the framework for data visualization using clustering algorithm. In this project, the DBSCAN algorithm was implemented using the number of confirmed and death cases. The processed data were scaled and the values were fit and normalized. The parameter used for this project was $eps = 0.1$, $min\_samples = 3$, and $metric = euclidean$. There are 4 clusters used to visualize which country belongs to which cluster. Fig. 4 shows the code snippet based on DBSCAN algorithm using Python.

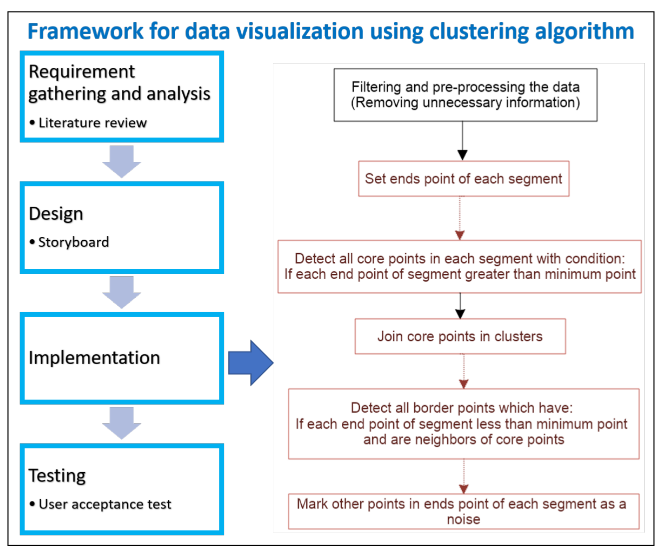

Fig. 3. Framework for data visualization using clustering algorithm (DBSCAN).

```
df3 = df3[['Confirmed','Deaths']]

df3 = preprocessing.scale(df3)#Data normalization

db=DBSCAN(eps=0.1,min_samples=3,metric='euclidean').fit(df3)
```

Fig. 4. Snippet code of DBSCAN algorithm using Python.

A. Phase 1: Requirement Gathering and Analysis

First and foremost, the appropriate approach to be used was selected throughout this requirement gathering process. Besides that, the gains and losses of every algorithm candidate was evaluated cautiously. Finally, every conceivable data source was investigated, with the most reliable being selected. As a result of the requirement gathering, it was found that the most suitable project topic is Data Visualization of COVID-19 Pandemic using a Density-Based Clustering Algorithm. Following that, data visualization and density-based clustering algorithm in the topic serve as markers of the project's focus. Furthermore, among other algorithms, the clustering method was chosen since the goal is to classify countries based on the severity of the infection, simulating a cluster of varying sizes. Finally, the data used were secondary data from the GitHub website. In the analysis phase, literature review was carried out. The methodology of the research was defined by examining articles dating back to 2018 up until 2021. Firstly, the problem statement centre around all of the undesired traits in the existing data visualization applications. Then, the objective and scope were acknowledged as well, which was to develop a web-based data visualization application using DBSCAN clustering algorithm. This project area covers the whole world since the pandemic affects us all.

1) Pre-processing of the data: The data obtained from the GitHub repository by the John Hopkins University, *https://github.com/CSSEGISandData/COVID-19* were cleaned before it can be used to gather useful information that can be used in the algorithm.

To clean the dataset, the Python programming language was used, using Jupyter Notebook. Firstly, the *.csv* file was read. The shape and data types of the data were described. The columns that were numeric and non-numeric were also identified to ease the upcoming processes. The result of the first block of code is that the dataset contained 81,437 lines of data with 62 columns. Out of the 62 columns, 5 columns were non-numeric. The columns in the dataset including the number of death and test that related to COVID-19. Fig. 5 shows the results generated from the dataset: Available columns in the data set. Then when the available columns were identified, the missing data were plotted and removed.

```
Length: 62, dtype: object
['day' 'hosp_patients' 'hosp_patients_per_million' 'icu_patients'
 'icu_patients_per_million' 'month' 'weekly_hosp_admissions'
 'weekly_hosp_admissions_per_million' 'weekly_icu_admissions'
 'weekly_icu_admissions_per_million' 'year' 'aged_65_older'
 'aged_70_older' 'cardiovasc_death_rate' 'diabetes_prevalence'
 'extreme_poverty' 'female_smokers' 'gdp_per_capita'
 'handwashing_facilities' 'hospital_beds_per_thousand'
 'human_development_index' 'life_expectancy' 'male_smokers' 'median_age'
 'new_cases' 'new_cases_per_million' 'new_cases_smoothed'
 'new_cases_smoothed_per_million' 'new_deaths' 'new_deaths_per_million'
 'new_deaths_smoothed' 'new_deaths_smoothed_per_million' 'new_tests'
 'new_tests_per_thousand' 'new_tests_smoothed'
 'new_tests_smoothed_per_thousand' 'new_vaccinations'
 'new_vaccinations_smoothed' 'new_vaccinations_smoothed_per_million'
 'people_fully_vaccinated' 'people_fully_vaccinated_per_hundred'
 'people_vaccinated' 'people_vaccinated_per_hundred' 'population'
 'population_density' 'positive_rate' 'reproduction_rate'
 'stringency_index' 'tests_per_case' 'total_cases'
 'total_cases_per_million' 'total_deaths' 'total_deaths_per_million'
 'total_tests' 'total_tests_per_thousand' 'total_vaccinations'
 'total_vaccinations_per_hundred']
['continent' 'date' 'iso_code' 'location' 'tests_units']
```

Fig. 5. Available columns in the data set.

B. Phase 2: Design

The flow of how the system operates should be obvious at this point of development, as this will decide what the system requires and does not require. As a result, Fig. 6 shows the system's flow which is depicted in a flowchart to make it easier to comprehend. The flow of the dashboard depends on whether the user changes the filters or not. As long as the user does not toggle any of the filters, the display will not change. Other than the flow of the system, the storyboard was also created in order to provide a guide when developing the system.

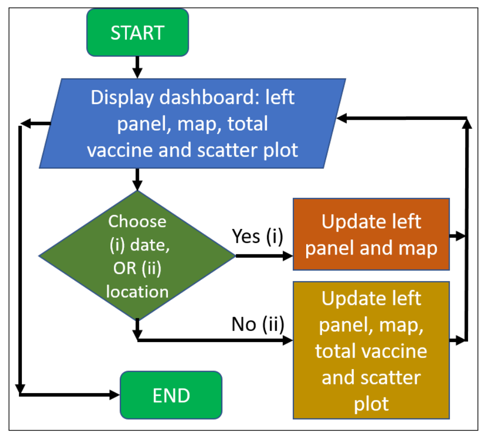

Fig. 6. Flowchart of the system.

C. Phase 3: Implementation

In the implementation phase, Jupyter Notebook was used to code the system in Python 3.

The system has two primary roles, as a:

1) dashboard to visualize the data of pandemic from all around the world. The DBSCAN algorithm used to detect and group any countries with similar death and case ratio together.
2) live Twitter feed; users can look at any recent updates regarding the COVID-19 pandemic.

Fig. 7 shows the screen-shot of the complete system. The system is only one page because a good dashboard is the one that can display everything at a glance.

There are two filters provided in this system: (i) a drop-down menu for the countries and (ii) a date picker. The dashboard displays a brief summary of the current pandemic situation based on the country and date chosen. By default, the system displays the data for Malaysia, on 4th July 2021. The reason is because the data in the system only goes back as far as 4th

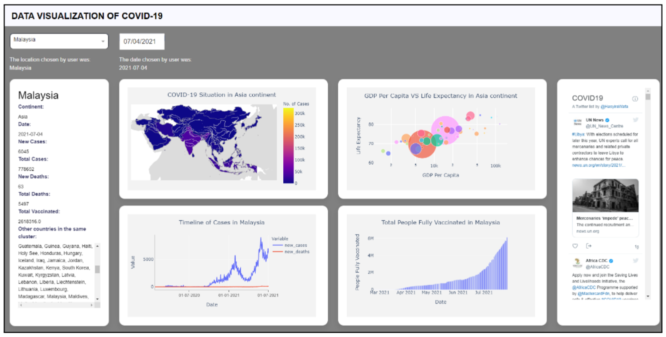

Fig. 7. Dashboard.

July 2021 which is the maximum date that can be selected. The data clustered using the DBSCAN algorithm to detect and group any countries with similar death and case ratio together.

Each of the clusters are represented by different colour. As a result, each country can be identified to one cluster each. The map represents each country with a different colour, according to its number of cases, referring to the colour scale. For example, in Fig. 8, it can be observed that most countries in the Asian continent have daily cases of less than 50,000.

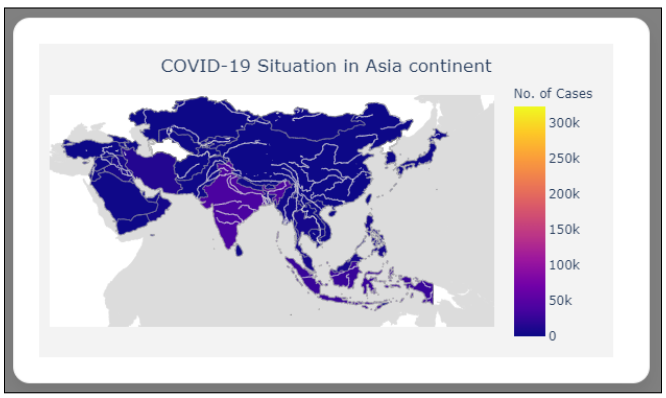

Fig. 8. COVID-19 map in Asian continent.

Fig. 9 portrays a simple line graph which has two different data on it: (i) number of new cases and (ii) number of new deaths.

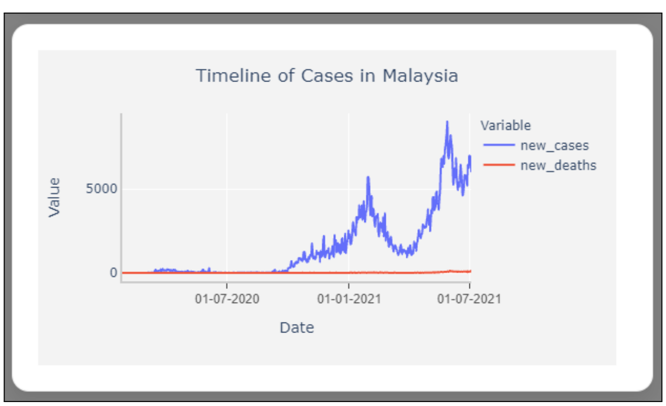

Fig. 9. Number of cases based on time-line.

Fig. 10 shows an example of the scatter plot for the Asian continent. There are different plots for different continents. The size of each bubble represents the population of the country. The bigger the bubble, the bigger the population. The scatter plot scatters the data with life expectancy on the y-axis and GDP per capita on its x-axis.

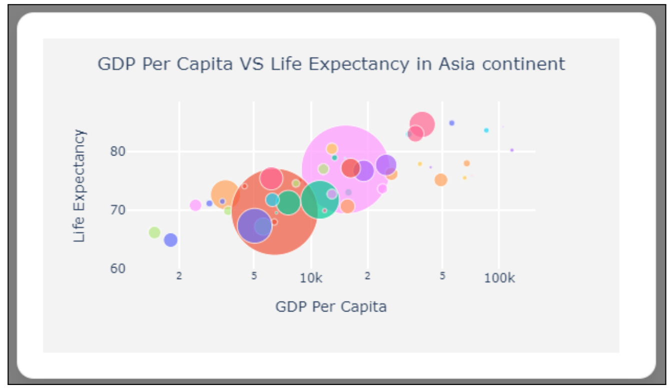

Fig. 10. Scatter plot for GDP Per Capita vs life expectancy in Asia.

Fig. 11 shows the example of individuals who are fully vaccinated in Malaysia. Based on the graph, more than 6,000,000 individuals have been vaccinated as of 28th July 2021.

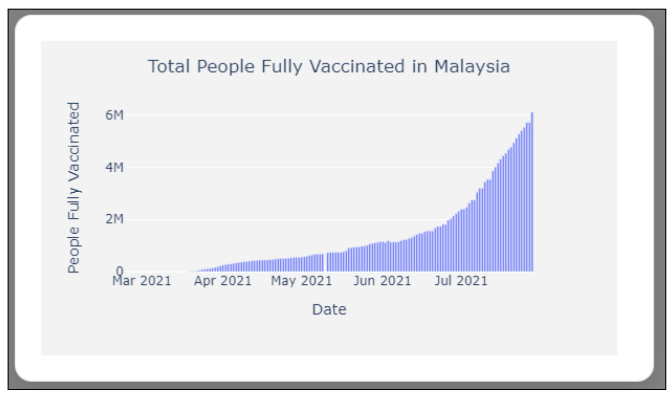

Fig. 11. Total of individuals that have been vaccinated in Malaysia.

D. Phase 4: Testing

At this phase, testing was conducted by the developer in order to ensure the study meets their requirement. There are two tests conducted by the developer: (i) Functionality test and (ii) Usability testing. The intended output was written down first in a (i) functionality test, and then the actual result was written down after the testing was completed. If the outcome did not meet the expectations, necessary improvements must be made.

In addition, (ii) usability testing questions were given to users in order to collect feedback. The users used then reacted on how they felt towards the system. The usability testing was carried out by 30 users. The age range of most of the respondents was from 18 to 24 years old.

TABLE I
ANALYSIS OF USABILITY TESTING QUESTIONS

	Usability testing questions	Results (%)
1.	The dashboard is easy to the eyes.	66.7
2.	The dashboard provides relevant data.	76.7
3.	The data are easily interpretable.	70.0
4.	The dashboard is interactive.	90.0
5.	You can find the data that you need.	86.7
6.	You can easily navigate through the dashboard.	80.0
7.	You can easily navigate through the dashboard.	83.3
8.	You would use the dashboard again.	53.3
9.	You would recommend this dashboard to your peers.	53.3

The analysis of usability testing questions used in this project is shown in Table I. Overall, respondents would recommend the dashboard to their peers.

IV. Implication, limitation, future work and conclusion

This project used a Python approach to develop a Dash application. A Density-Based Spatial Clustering of Applications with Noise algorithm was used to classify countries into clusters based on its number of cases and number of deaths.

There are four main visualizations on the web application: (i) a choropleth map with the daily cases of each country represented by its respective colour according to the colour scale, (ii) a scatter plot which visualizes all of the countries into bubbles of sizes based on its population, (iii) a line chart with the daily number of new cases and deaths, and (iv) a visualization of the vaccination rate.

Interestingly, this project would benefit not only policy-makers, but public, government, healthcare providers, and businesses. Due to the nature of the work, policy-makers are dealing with unprecedented amount of data to aid with the decision-making process as they have to provide immediate guidelines and communicate the information effectively to the citizens, especially when dealing with the COVID-19 pandemic. For example, the guideline produced from the data visualization can be utilized to reduce the strict confinement measures [14].

Furthermore, as for the Ministry of Education (MOE), they can also fully utilize the dashboard in making decisions whether or not schools or universities should continue with online and distance learning (ODL). This is because it is crucial to know the most current situation of the pandemic in order to reduce the transmission of the virus in schools [15].

However, there are a few notable limitations of the application: (i) the dashboard does not contain live data, the data only goes up to 4th July 2021. Other than that, (ii) the default display is for Malaysia instead of automated based on the location of the user.

In conclusion, future research should focus on adding more consoles to the web application such as automatic location detection, utilizing real time data (E.g. data from MySejahtera application), using cookies to save user history, adding a notification console, and other methods to make the process faster since we have to deal with a big amount of data.

Acknowledgement

This project was supported by Universiti Teknologi MARA Malacca Branch (Jasin Campus) under the TEJA Internal Grant Scheme 2021 (**GDT2021/1-26**).

Authors would like to thank Center of Vision and Algorithms Analytics (C-VAA) and Information Technology for Organizations (ITFO) Research Groups for the supports throughout this research.

This project has received the ”Industrial Panel Award” at the Final Year Project Exhibition Faculty of Computer & Mathematical Sciences (FSKM) Malacca Branch (Jasin Campus) (20212).

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